

Ex: let  $\langle , \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$  be given by  $\langle x, y \rangle := x_1 y_1 + 4x_2 y_2$   
for basis  $[b_1, b_2]$ , other basis  $[c_1, c_2]$

What is the formula for  $\langle , \rangle$  in  $C$  basis?

$$\begin{aligned} c_1 &= b_1 & x &= \tilde{x}_1 c_1 + \tilde{x}_2 c_2 = \tilde{x}_1 b_1 + \frac{\tilde{x}_2}{2} b_2 \\ c_2 &= \frac{1}{2} b_2 & y &= \tilde{y}_1 c_1 + \tilde{y}_2 c_2 = \tilde{y}_1 b_1 + \frac{\tilde{y}_2}{2} b_2 \end{aligned}$$

In  $C$ -basis  $\langle x, y \rangle = \tilde{x}_1 \tilde{y}_1 + 4 \frac{\tilde{x}_2}{2} \frac{\tilde{y}_2}{2} = \tilde{x}_1 \tilde{y}_1 + \tilde{x}_2 \tilde{y}_2 = x \cdot y$

in the  $b$ -basis  $\langle , \rangle$  is "unusual" but in the  $C$ -basis  $\langle , \rangle$  is just the dot product

---

Properties of the dot product on  $\mathbb{R}^n$

- $(1, 0, \dots, 0) \cdot (1, 0, \dots, 0) = 1$  if  $v$  is a basis vector  $v \cdot v = 1$
  - $(1, 0, \dots, 0) \cdot (0, 1, 0, \dots, 0) = 0$  if  $v$  and  $w$  are 2 diff elements of the basis then  $v \cdot w = 0$
- 

Def: given  $v$  and an inner product  $\langle , \rangle$ , a basis is called orthonormal if

- 1)  $\langle b_i, b_i \rangle = 1 \quad \forall i$
- 2)  $\langle b_i, b_j \rangle = 0$  for  $i \neq j$

So our example in the  $b$ -basis is Not orthonormal but it is orthonormal in the  $C$ -basis

---

Ex:  $\mathbb{R}^2$  and the dot product,  $v = (x, y)$   $\langle v, v \rangle = x^2 + y^2$

if  $\langle v, v \rangle = 1$  this is all points dist 1 from  $(0, 0)$

if  $\langle v, v \rangle = k$  this is all points dist  $\sqrt{k}$  from  $(0, 0)$

$\sqrt{\langle v, v \rangle} = \sqrt{x^2 + y^2}$  is the "length" of the vector  $v$ .

Def: given an inner product  $\langle, \rangle$  on  $V$ , the length of a vector  $v$  is given by  $\|v\| := \sqrt{\langle v, v \rangle}$

Ex: in  $\mathbb{R}^2$  w/  $\langle x, y \rangle := x_1 y_1 + 4x_2 y_2$

what is the length of  $(1, 1)$ ? what is the length of  $(0, 0)$

$$\langle (1, 1), (1, 1) \rangle = 1 + 4 = 5, \quad \langle (0, 0), (0, 0) \rangle = 0$$

$$\|(1, 1)\| = \sqrt{5}$$

$$\|(0, 0)\| = 0$$

Lem: in any vector space w/ any inner product  $\|0\| = 0$

Pf:  $\langle 0, 0 \rangle = \langle 0 \cdot 0, 0 \rangle = 0 \langle 0, 0 \rangle = 0$

↑  
0 scalar times 0 vector

length is an inner product dependent concept

Def: given a vector space and an inner product  $\langle, \rangle$

the angle between  $v, w$  is defined to be the smallest

$$\theta \text{ s.t. } \cos \theta = \frac{\langle v, w \rangle}{\|v\| \|w\|} = \left\langle \frac{v}{\|v\|}, \frac{w}{\|w\|} \right\rangle$$

note: in  $\mathbb{R}^n$  w/ dot product, this is our intuitive notion of angle

Def:  $\frac{v}{\|v\|}$  is the unit vector in the same dir. as  $v$   
the normalization of  $v$

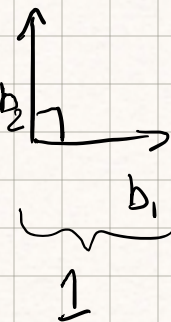
Ex:  $[b_1, b_2], [c_1, c_2]$   $c_1 = b_1, c_2 = \frac{1}{2}b_2$

if  $x = x_1 b_1 + x_2 b_2$   $y = y_1 b_1 + y_2 b_2$  then  $\langle x, y \rangle = x_1 y_1 + 4x_2 y_2$

what is a geometrically accurate picture of  $\mathbb{R}^2$  w/ basis  $B$



$$\cos \theta = \frac{\langle b_1, b_2 \rangle}{\|b_1\| \|b_2\|} = \frac{\cancel{(1(0) + 4(0)(1))}^0}{\|b_1\| \|b_2\|} \quad \cos \theta = 0 \quad \theta = \frac{\pi}{2}$$



$$\|b_1\| = \sqrt{\langle b_1, b_1 \rangle} = 1$$

$$\|b_2\| = \sqrt{\langle b_2, b_2 \rangle} = \sqrt{4} = 2$$

$$\|c_1\| = 1$$

$$\|c_2\| = 1 = \sqrt{\langle c_2, c_2 \rangle} = \sqrt{\langle \frac{1}{2}b_2, \frac{1}{2}b_2 \rangle} = \sqrt{\frac{1}{4}\langle b_2, b_2 \rangle} = 1$$

